RAG + Vector Database Lab Report

Project: cursor-rag-lab-sam (Cursor RAG Lab Sam)
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Account: sammycloud975@gmail.com
Stack: Python · Vertex AI embeddings · BigQuery vector search · Gemini
Repo path: rag-lab/

Goal completed: Implement Retrieval-Augmented Generation (RAG) using a vector database on Google Cloud, with a runnable demo that retrieves company-policy chunks from BigQuery and answers with Gemini.

1. Architecture

```
Sample docs → chunk → Vertex text-embedding-005 (768-dim)
                                ↓
                 BigQuery rag_lab.doc_chunks  ←— vector store
                                ↑
User question → same embedding model → VECTOR_SEARCH (cosine, top-k)
                                ↓
                 Gemini 2.5 Flash answers from retrieved context only
                                ↓
                 BigQuery rag_lab.query_log stores each Q&A run
```

2. What was created in Google Cloud

Resource	Details	Status
GCP project	ID `cursor-rag-lab-sam`, number 84704792466, under org 1058021731899	**Created**
Billing	Linked to billing account 01F59D-CE68A5-998B16	**Enabled**
API: Vertex AI	`aiplatform.googleapis.com`	**Enabled**
API: BigQuery	`bigquery.googleapis.com`	**Enabled**
Dataset	`rag_lab` (location US)	**Created**
Table `doc_chunks`	Vector store: id, doc_id, content, embedding[], created_at	**8 rows**
Table `query_log`	Demo audit log: question, answer, retrieved_ids, distances, models	**Created**

Embeddings model	Vertex `text-embedding-005` (768 floats per chunk)	**Used**
Generation model	Vertex `gemini-2.5-flash`	**Used**

Note: A BigQuery IVF vector index was attempted but requires ≥5000 rows. This lab correctly uses `VECTOR_SEARCH` directly on the small corpus (still a valid vector-database workflow).

3. Document corpus ingested

Source file	Topics
`data/sample_docs/it_faq.txt`	Password reset, VPN, laptop setup
`data/sample_docs/hr_handbook.txt`	Vacation, parental leave, remote work
`data/sample_docs/security_policy.txt`	Phishing, MFA, data classification
`data/sample_docs/expense_policy.txt`	Expenses, travel, software subscriptions

Documents are chunked (~500 characters), embedded, and loaded into `doc_chunks` (8 chunks total).

4. Code delivered in the repo

Module	Role
`src/config.py`	Loads project settings from `.env`
`src/embed.py`	Vertex embedding helper / CLI
`src/setup_bq.py`	Creates dataset + tables
`src/ingest.py`	Chunk → embed → BigQuery load
`src/retrieve.py`	BigQuery `VECTOR_SEARCH` with similarity filter
`src/generate.py`	Gemini answer from retrieved context
`src/log_query.py`	Writes runs to `query_log`
`src/rag_pipeline.py`	Single-question end-to-end RAG
`src/demo.py`	Multi-question showcase demo
`lessons/` + `SHOWCASE.md`	Tutor notes and presentation guide

5. Example demo results (already run in cloud)

Question	Top source	Answer summary
How do I reset my password?	it_faq	Okta reset URL; 30-min link; Slack #it-help if locked out
How many vacation days…?	hr_handbook	20 paid days per year
What if I get a phishing email?	security_policy	Don't click; report to security@ / Report Phish
Daily meal limit while traveling?	expense_policy	$75/day without pre-approval
Remote three days a week?	hr_handbook	Yes, up to 3 days with manager approval

6. How to see the vector database in Cloud Console

1. Open project: `https://console.cloud.google.com/home/dashboard?project=cursor-rag-lab-sam`
2. Open BigQuery: `https://console.cloud.google.com/bigquery?project=cursor-rag-lab-sam`
3. Left explorer → **cursor-rag-lab-sam** → **rag_lab** → **doc_chunks**
4. Click **Preview** to see chunk text and the `embedding` array (768 numbers) — this is the vector DB
5. Open **query_log** → **Preview** to see logged RAG Q&A runs

```
-- Optional SQL in BigQuery editor:
SELECT id, doc_id, LENGTH(content) AS chars, ARRAY_LENGTH(embedding) AS dims
FROM `cursor-rag-lab-sam.rag_lab.doc_chunks`
ORDER BY doc_id, id;
```

7. How to test RAG (commands)

```
cd rag-lab
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
cp .env.example .env               # ensure PROJECT_ID=cursor-rag-lab-sam

# (Re)build tables if needed, then ingest + demo
python -m src.setup_bq
python -m src.ingest
python -m src.demo

# Or ask one question:
python -m src.rag_pipeline "How do I reset my password?"
python -m src.retrieve "What should I do if I get a phishing email?"
```

What "working" looks like: the CLI prints retrieved chunk sources with similarity scores, then a Gemini answer that cites those chunks (e.g. `[1]`). A new row appears in `query_log`.

8. Console quick links

- Project home: https://console.cloud.google.com/home/dashboard?project=cursor-rag-lab-sam
- BigQuery: https://console.cloud.google.com/bigquery?project=cursor-rag-lab-sam
- Enabled APIs: https://console.cloud.google.com/apis/dashboard?project=cursor-rag-lab-sam

Generated for the gcp-rag-vector-lab coursework materials. No Cloud Run / public web app was deployed; the deliverable is the BigQuery vector store plus the Python RAG pipeline and demo.